★TOPIC:- HARDWARE

Module_1-Day_3:

❖ Tasks:

1. **Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.**

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❖ Comparison between DDR3, DDR4, and DDR5 RAM:

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
Speed	800–2133 MT/s	1600–3200 MT/s	4800–8800+ MT/s
Voltage	1.5V (typical)	1.2V	1.1V
Performance	Low	Better	Highest
Power Consumption	Higher	Lower	Lowest voltage; improved efficiency
Typical Use Cases	Older desktop PCs and laptops	Many older and mid-range PCs	Modern gaming PCs, workstations, and high-performance computers

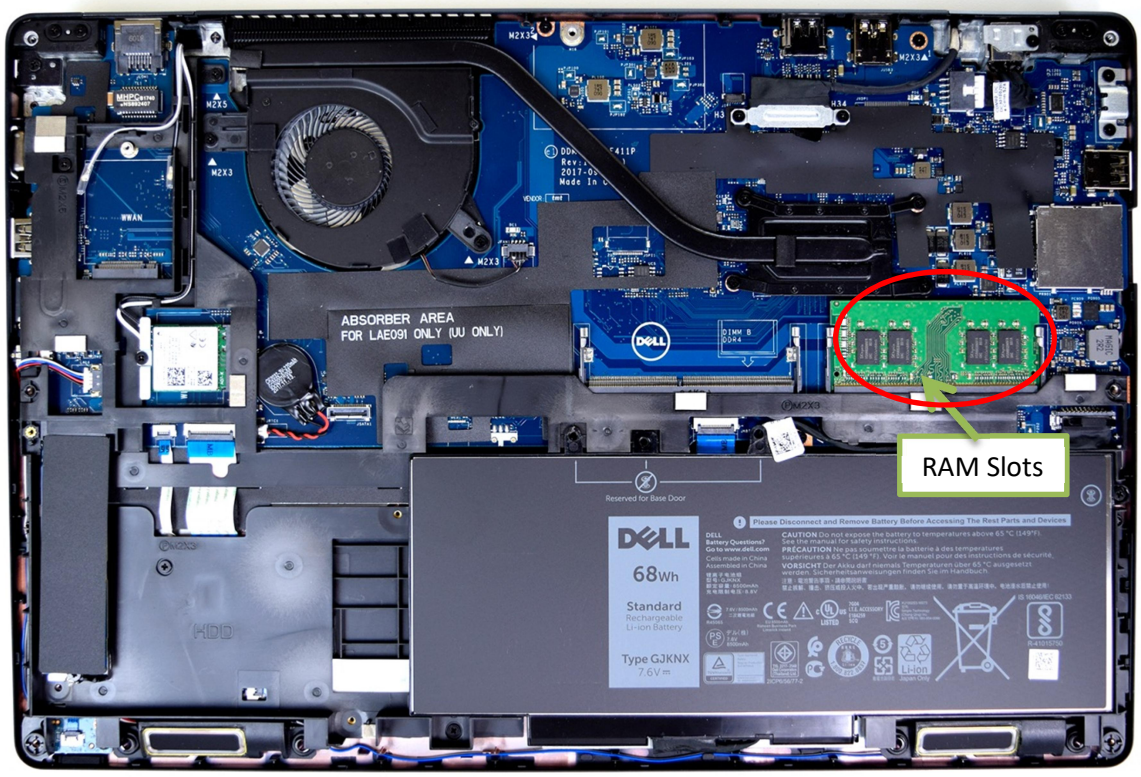
2. **Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.**

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❖ RAM Removal and Reinstallation Steps to perform:

(1) Switch off the laptop and disconnect the charger and all connected devices.
(2) Open the laptop's back cover using the correct screwdriver. Consult the service manual first.
(3) Disconnect the internal battery before touching the RAM.
(4) Locate the RAM module and its slot on the motherboard.
(5) Carefully push the two side clips outward. The RAM will lift up; slide it out gently.
(6) Take a clear photo of the RAM module and its slot.
(7) Align the notch in the RAM with the slot and insert it at an angle. Press it down gently until the clips lock into place.
(8) Reconnect the battery, close the cover, and power on the laptop.
(9) Confirm that the laptop starts normally and that the installed RAM is detected in Windows Task Manager under Performance → Memory.



3. **List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.**

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❖ Difference between HDD, SSD, and NVMe

Feature	HDD	SSD	NVMe
Speed	Slow	Fast	Very fast
Technology	Uses spinning disks	Uses flash memory chips	Uses flash memory with a faster connection
Cost	Cheap	More expensive than HDD	Usually more expensive
Storage Capacity	Available in large capacities	Available in different capacities	Available in different capacities
Example / Use Case	Storing movies and photos	Opening Windows and Google Chrome quickly	Loading heavy games and large software quickly

4. **Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.**

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❖ Storage Drive Removal and Reinstallation Steps to perform:

(1) Shut down the computer and disconnect the power cable and all peripherals.
(2) Open the laptop or desktop case using the appropriate screwdriver and service instructions.
(3) Locate the HDD or SSD and identify its connector:

- ★ SATA: A wider data/power connection used by many 2.5-inch SSDs and HDDs.
- ★ M.2: A small, flat connector used by compact SSDs.
- ★ NVMe: A storage protocol commonly used by M.2 SSDs; it is not a separate physical connector type.

(4) Carefully remove the drive according to the device's service manual.
(5) Take a clear photograph of the drive and its connector.
(6) Reinstall the drive, secure it properly, and reconnect any cables.
(7) Close the case, reconnect power, and start the computer.
(8) Confirm that the drive is recognized in BIOS/UEFI or the operating system. In Windows, check Disk Management or Device Manager.

Storage Drive Removal and Reinstallation (HDD/SSD)

1 Shut Down and Disconnect



- Shut down the computer.
- Disconnect the power cable and all peripherals.

2 Open the Laptop/Case



- Remove the screws from the bottom panel using a suitable screwdriver.
- Carefully open the case (follow your model's service manual).

3 Locate the Storage Drive



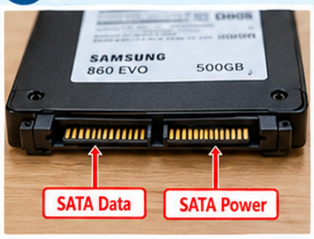
- Find the HDD or SSD in the laptop.
- In this example, a 2.5-inch SSD is used.

4 Remove the Drive



- Disconnect the SATA connector (if any).
- Remove the screws or bracket.
- Gently take out the drive.

5 Photo of the Connector Type



- This drive uses a SATA connector (wide data + power connection).

6 Reinstall the Drive



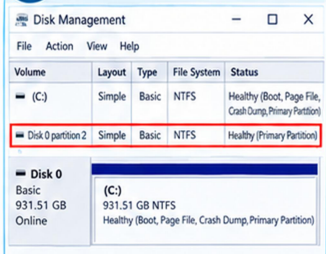
- Place the drive back in the slot.
- Reconnect the SATA connector (if any).
- Secure it with screws or bracket.

7 Close the Laptop/Case



- Put the bottom panel back.
- Tighten all screws.
- Reconnect the power cable and peripherals.

8 Check if the Drive is Recognized



- Open Disk Management (Windows).
- Confirm the drive is detected and accessible.

Connector Types Reference

SATA



2.5-inch SSD/HDD

Wide data + power connector

M.2 (SATA)



M.2 SSD (SATA)

Small, flat connector
(uses SATA protocol)

M.2 (NVMe)



M.2 SSD (NVMe)

Same physical slot as SATA
but uses NVMe protocol
(faster performance)